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Illinois Hospital CHNA and County IPLAN Comparison

1. Project Background and Purpose

- Compare hospital Community Health Needs Assessments (CHNAs) with county health department IPLAN priorities across Illinois.
- Examine whether hospital and county health priorities align and how those patterns relate to population demographics and healthcare resources.
- Identify potential geographic disparities and propose a future GIS tool to visualize the findings.

2. Work Completed: Data Collection and Preparation

- Compiled hospital CHNA priorities and county health department IPLAN priorities.
- Created separate datasets and standardized health priority categories to support comparisons.
- Collected available assessment information and documented missing or unavailable plans.
- Prepared the datasets for geographic matching and comparison by county.

3. Proposed Analysis: Comparing CHNA and IPLAN Priorities

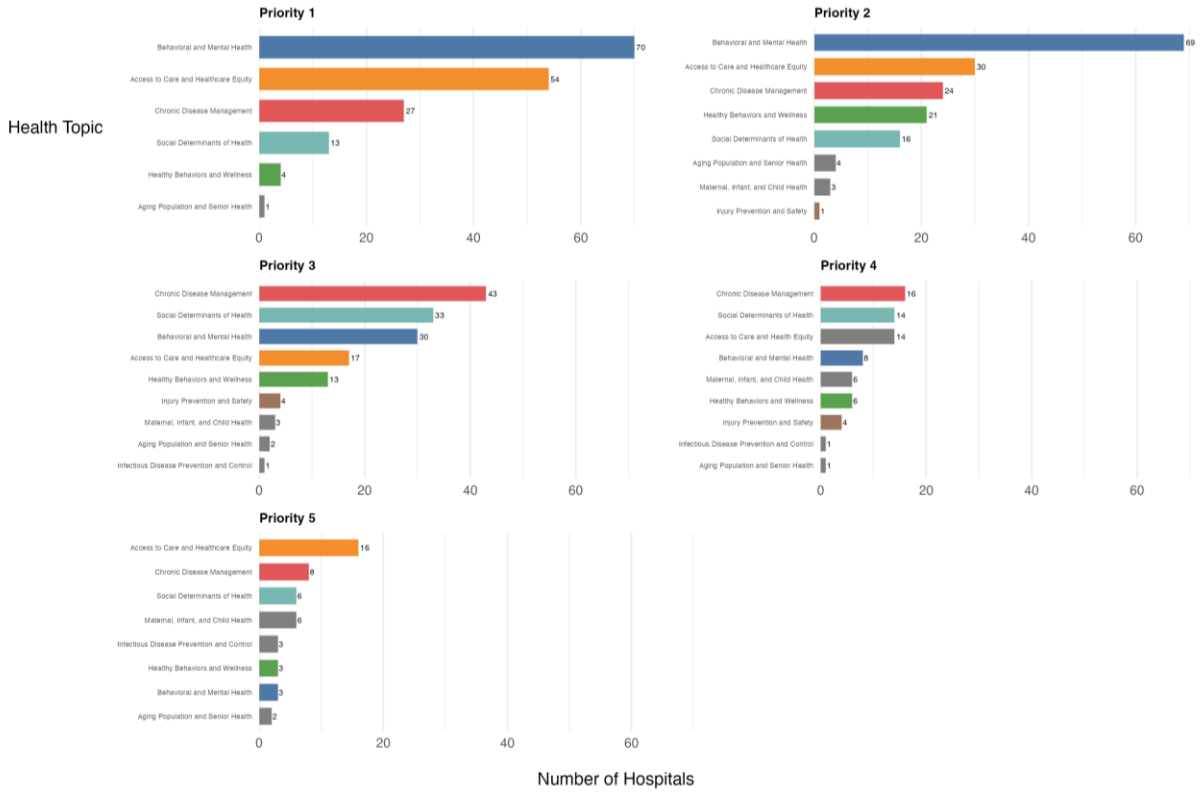
- Match hospital CHNA priorities with the corresponding county IPLAN priorities.
- Identify shared priorities, priorities unique to hospitals, and priorities unique to health departments.
- Calculate the percentage of priorities that overlap between the two sources.
- Compare priority alignment across counties and examine differences among multiple hospitals within the same county.
- **Proposed calculations:**
    - County priority coverage: Percentage of county IPLAN priorities also identified in hospital CHNAs.
    - Hospital priority coverage: Percentage of hospital CHNA priorities also identified in the county IPLAN.
    - Priority overlap: Percentage of all distinct priorities identified by either source that appear in both.

4. Proposed Population and Demographic Analysis

- Add U.S. Census Bureau population estimates and demographic data for each county.
- Include population size, poverty, and rurality.
- Compare population demographics with hospital availability and health priority patterns.

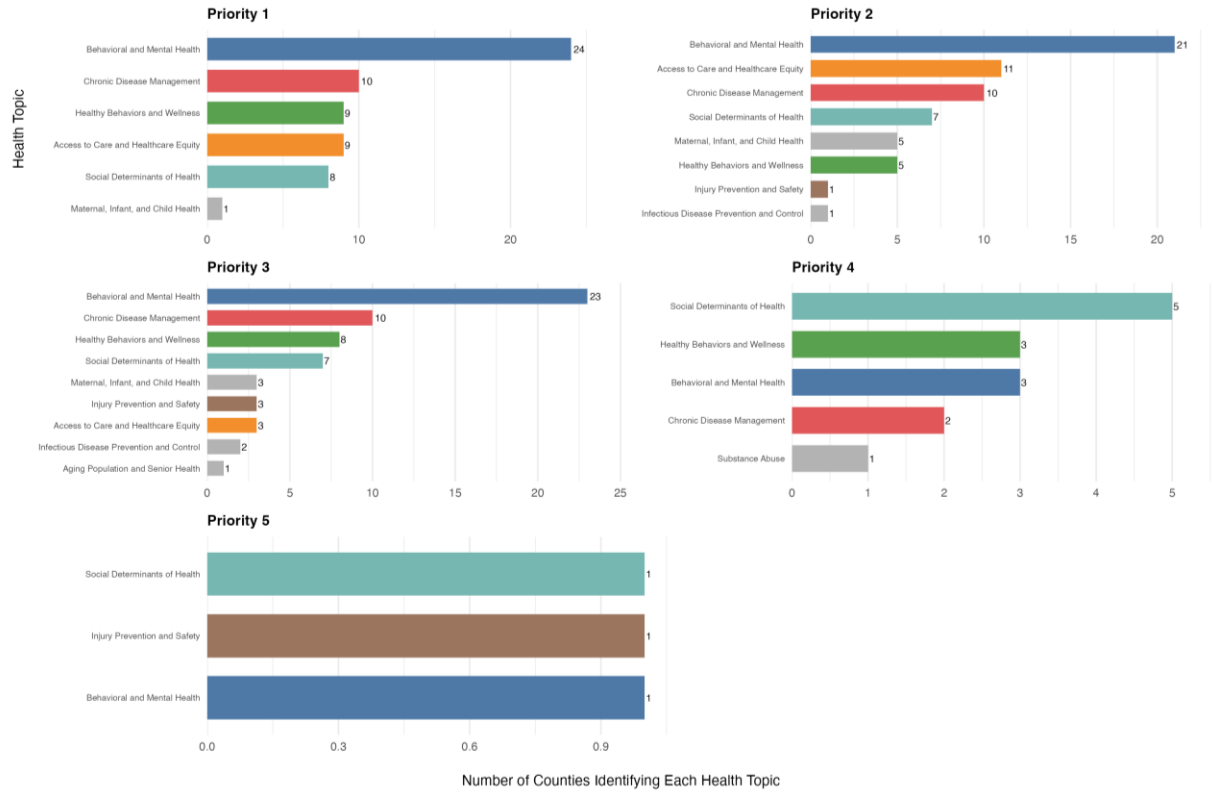
- Examine whether counties with greater socioeconomic disadvantage show different priority alignment or potential gaps in healthcare resources.
- **Proposed calculations**
    - Hospitals per 100,000 residents: Number of hospitals ÷ County population) × 100,000
    - Priority overlap percentage: (Number of shared priorities ÷ Total number of distinct priorities across both sources) × 100
    - Compare demographic percentages and hospital rates across counties to identify patterns that may warrant further investigation.

Distribution of Community Health Needs Assessment Health Priorities Across Illinois Hospitals



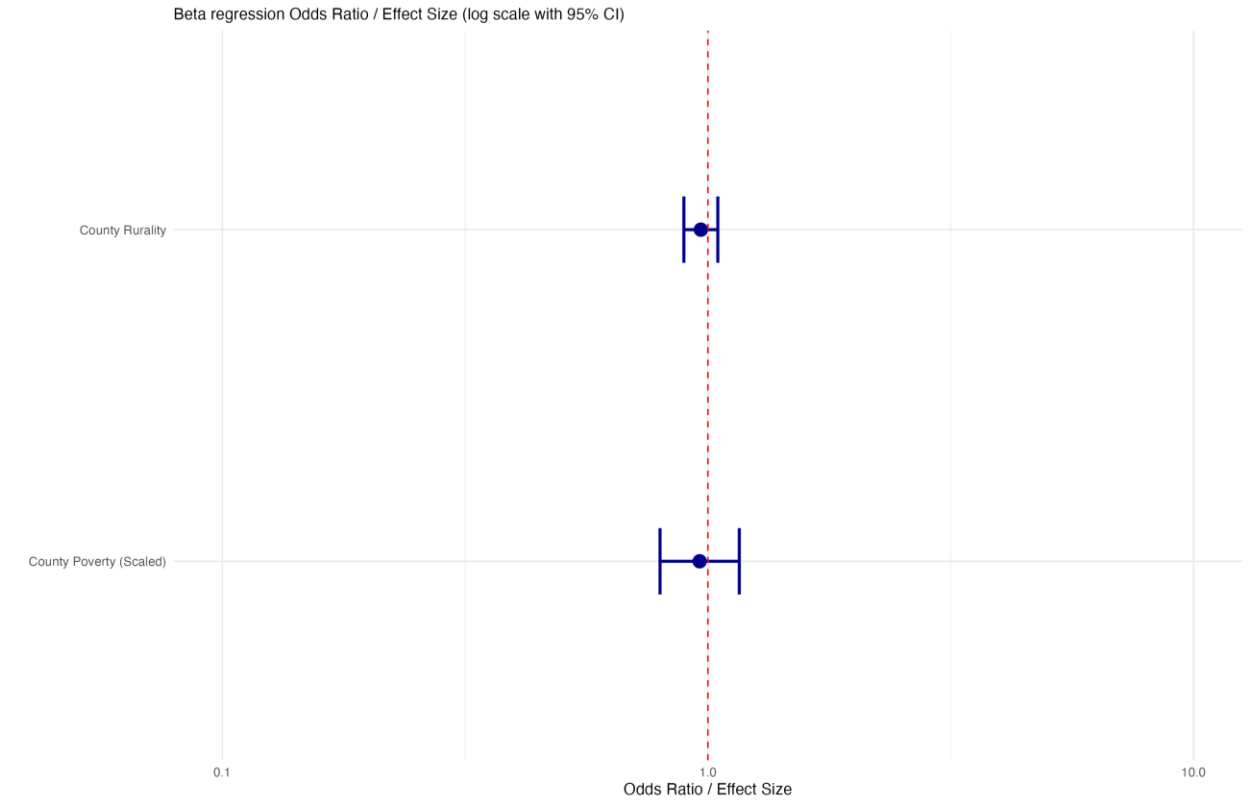
Hospital Community Health Needs Assessments (CHNAs) overwhelmingly rank Behavioral and Mental Health as Priority 1 and Priority 2. However, hospitals place a significantly higher initial priority on Access to Care and Healthcare Equity, making it the second most common Priority 1, compared to county public health plans. This graph illustrates key institutional health system priorities; while hospitals align closely with counties on mental health needs, their strong initial emphasis on access to care reflects urgent clinical and operational challenges, such as emergency department utilization and insurance coverage barriers.

Distribution of Community Health Needs Assessment Health Priorities Across Illinois Counties



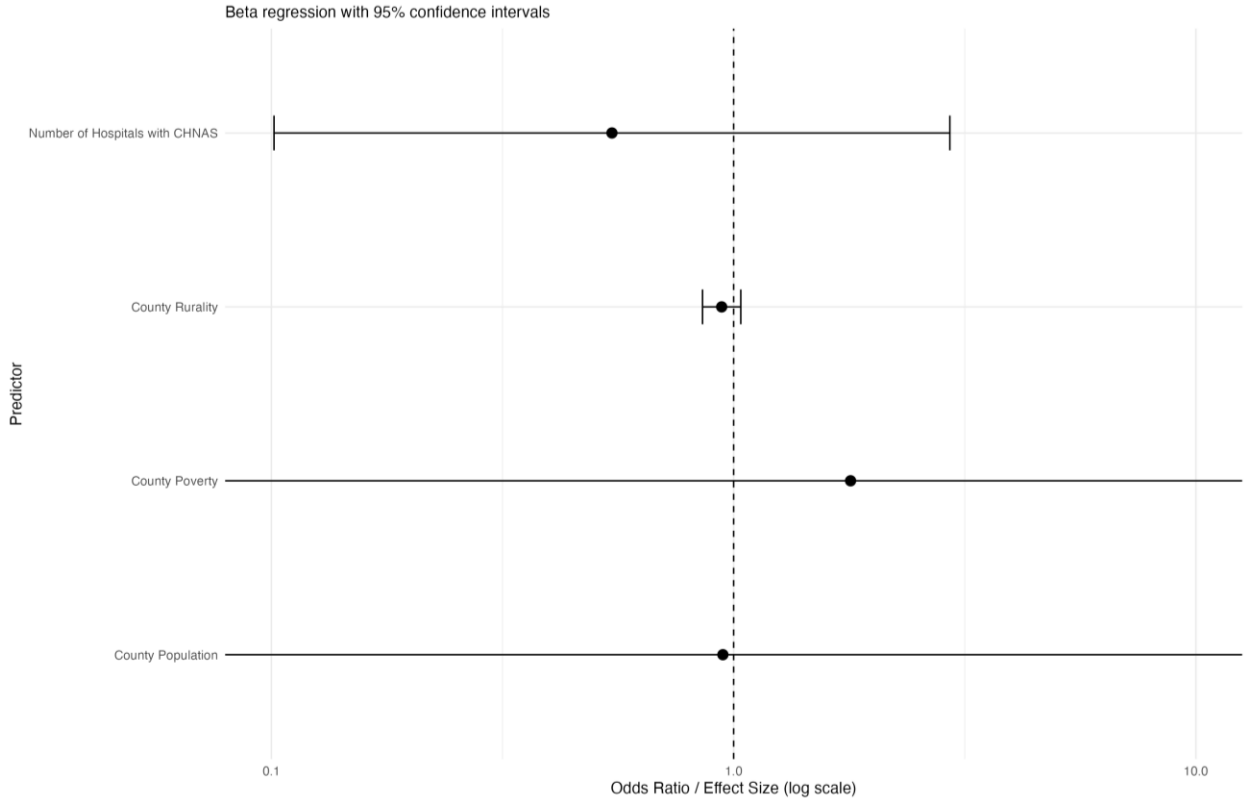
Across top-ranked priorities (Priority 1 through 3), Behavioral and Mental Health is consistently the most frequently identified health need by Illinois counties, followed by Chronic Disease Management, Healthy Behaviors and Wellness, and Access to Care and Healthcare Equity. Secondary and tertiary priorities (Priority 4 and 5) shift toward broader systemic factors like Social Determinants of Health. This visualization highlights a strong public health consensus across Illinois, demonstrating that mental health and chronic disease management form the primary core of county health assessment strategies statewide.

Factors Associated with County Priority Overlap



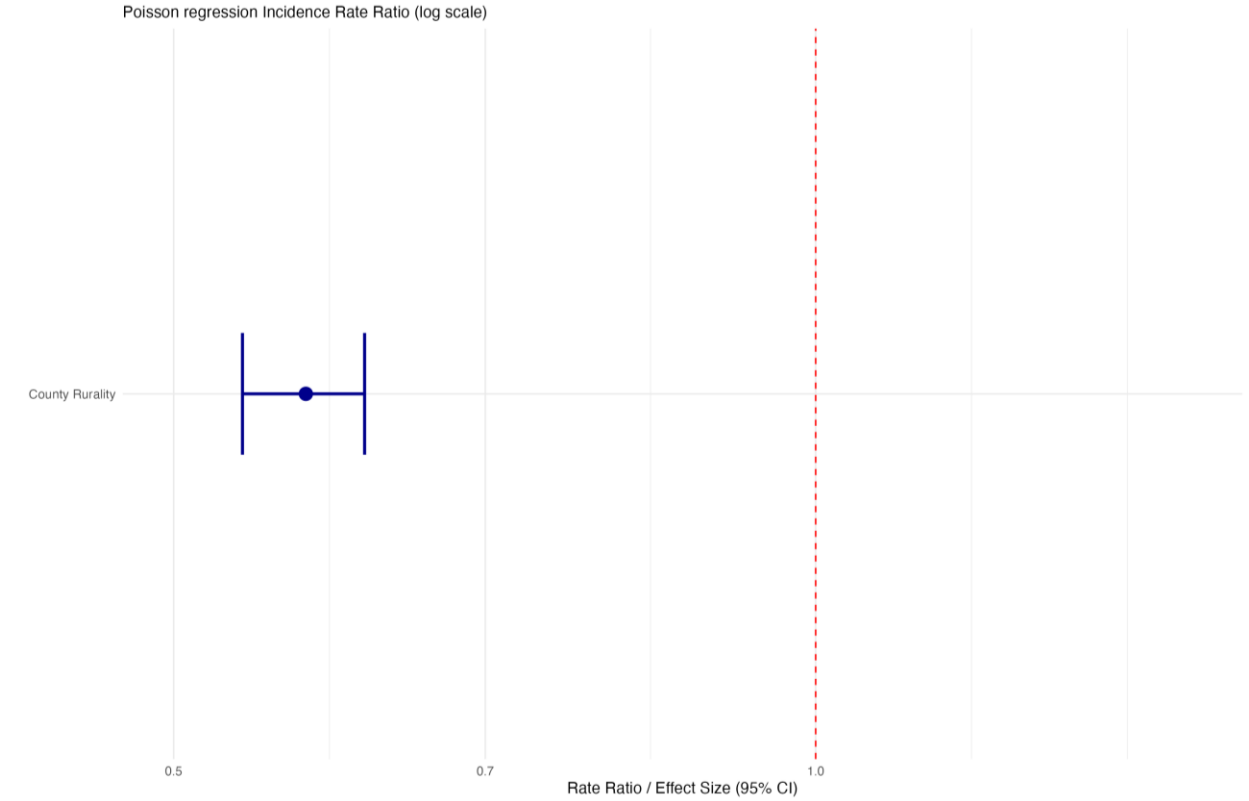
A refined beta regression specifically evaluating County Rurality and Scaled County Poverty shows point estimates for both variables centered almost directly at 1.0, with narrow confidence intervals clearly spanning the null line. This graph statistically confirms that neither rurality nor poverty scale directly influences the degree of priority overlap between entities. Consequently, priority alignment is shown to be independent of these demographic metrics, emphasizing that alignment depends on context-specific strategies rather than county demographics.

Factors Associated with County–Hospital Priority Overlap



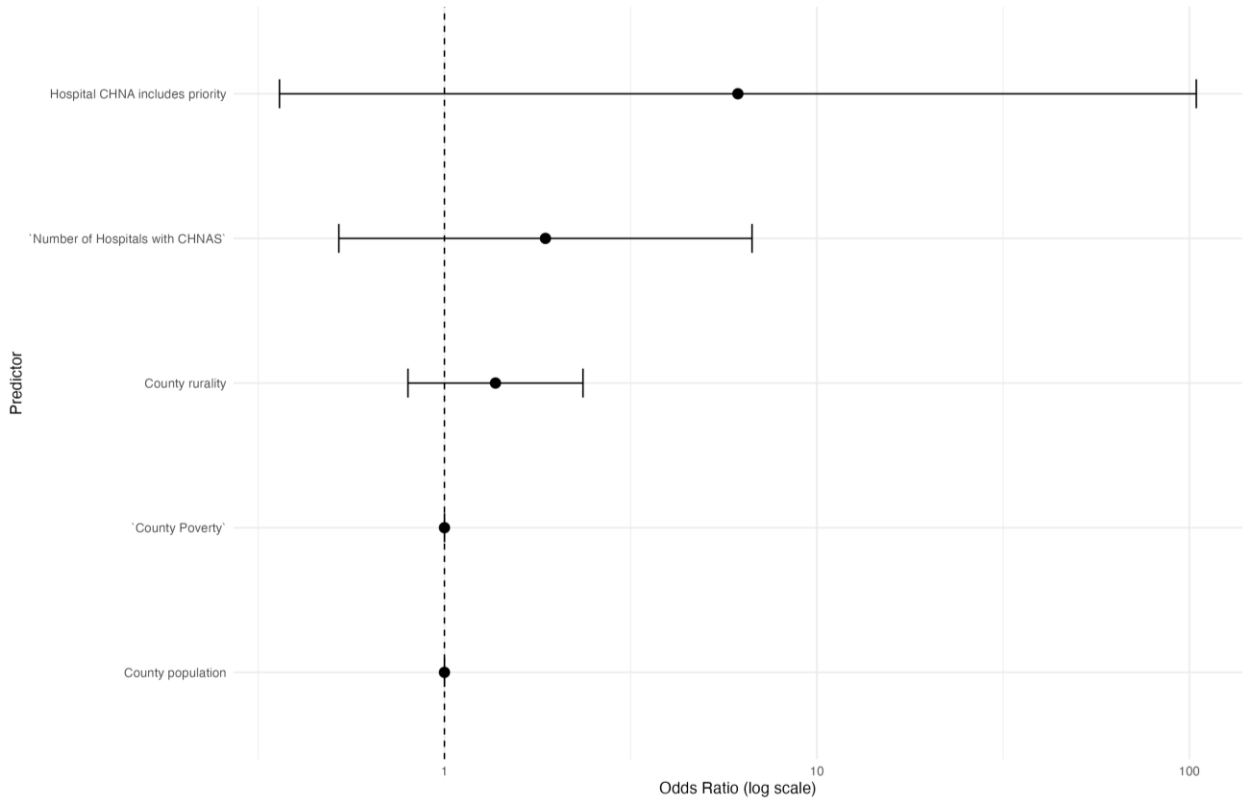
Results from a beta regression examining the alignment between county and hospital priorities show that county poverty has a positive point estimate toward greater overlap, whereas county rurality and the number of hospitals lie slightly below 1.0. However, the 95% confidence intervals across all predictors are exceptionally wide and cross 1.0. This suggests that broad structural county characteristics like poverty or rurality do not reliably predict how closely hospital and county health priorities align, pointing instead toward localized collaboration dynamics as the key driver of alignment.

Effect of County Rurality on Hospital Density



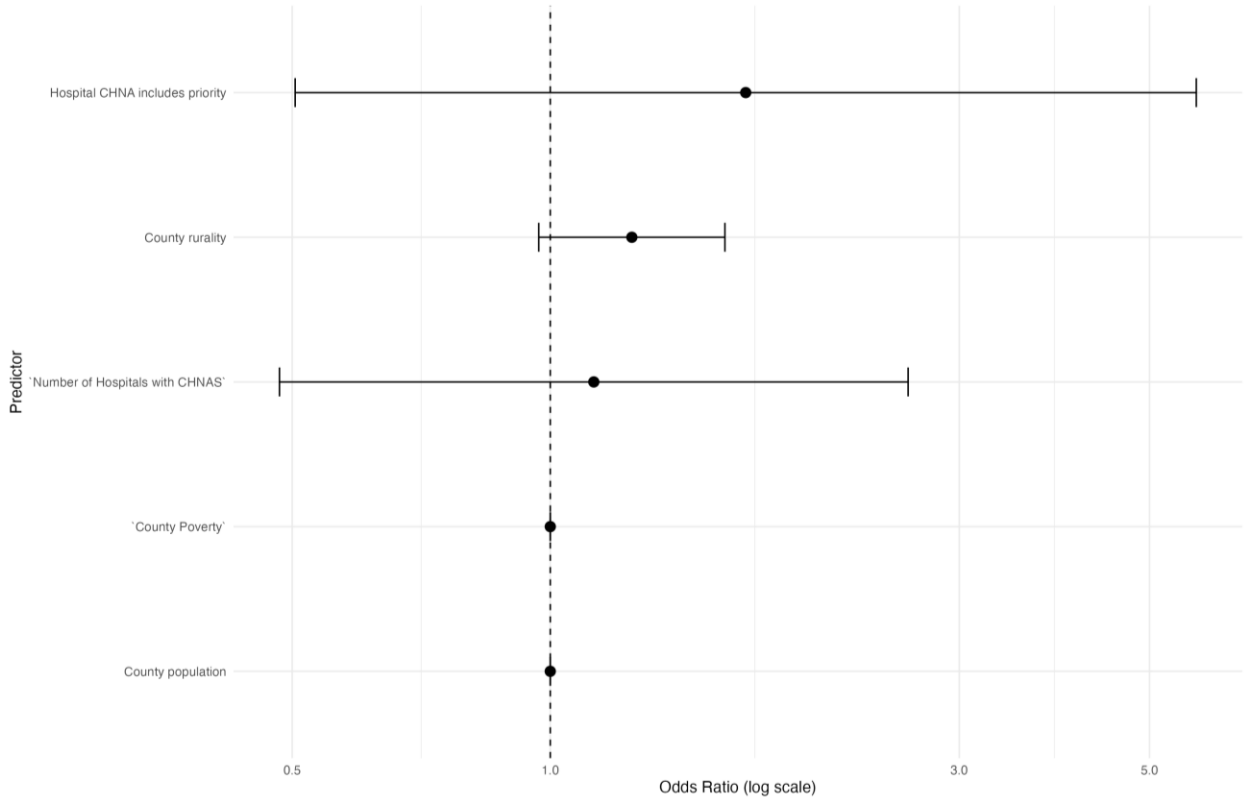
A Poisson regression incidence rate ratio (IRR) plot shows that County Rurality has an effect size significantly below 1.0 (IRR ≈ 0.6), with its entire 95% confidence interval falling well to the left of the reference line. This graph statistically establishes a robust negative association between rurality and hospital density. Because more rural counties have significantly fewer hospitals per capita and geographic area, this finding highlights a key structural access barrier that complicates cross-sector health planning and resource allocation.

Factors Associated with Behavioral and Mental Health Priorities



This forest plot displays odds ratios on a log scale to show factors predicting whether behavioral health is prioritized. The strongest positive predictor is whether local hospital CHNAs include it as a priority. County rurality and the number of hospitals with CHNAs also trend positive, though their 95% confidence intervals cross or approach 1.0, while county poverty and population show odds ratios centered at 1.0. These findings reveal that institutional alignment—specifically a local hospital's focus—is the strongest driver of whether a county formally prioritizes behavioral health, far outweighing baseline demographic factors like population size or poverty.

Factors Associated with Chronic Disease Management Priorities



In evaluating chronic disease management, inclusion in a hospital CHNA demonstrates a strong positive point estimate, albeit with wide confidence intervals. County rurality displays a modest positive association, whereas the number of hospitals, county poverty, and county population all center around the null reference line of 1.0. This indicates that local hospital prioritization remains an influential factor in driving county-level chronic disease initiatives, while broader sociodemographic predictors alone do not strongly dictate whether chronic disease becomes a designated public health priority.

Future Directions: GIS Spatial Mapping Tool

Objective: Develop an interactive GIS mapping tool to spatially visualize and evaluate the geographic distribution of hospital Community Health Needs Assessments (CHNAs) alongside county IPLAN priorities across Illinois.

Application: Map hospital locations relative to county boundaries to visually identify regional health priority clusters, highlight areas of high versus low priority overlap, and uncover spatial healthcare resource disparities (such as rural hospital deserts).

Impact: Provide public health leaders and researchers with a visual interface to better target cross-sector resource allocation and collaborative health interventions.